Riccardo Visconti

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PROFILE

A motivated Computational Science Master's student at USI, holding a Bachelor's Degree in Mathematics and a solid foundation from a scientific high school. Seeking a part-time position or internship to apply analytical, quantitative, and problem-solving skills in a practical context, such as data analysis, machine learning, or complex problem-solving. Flexible, meticulous, and recognized for a strong ability to learn quickly.

EDUCATION

Master in Computational Science — Feb 2026 – Present
USI (Università della Svizzera italiana) — *Lugano, Switzerland*

Bachelor's Degree in Mathematics — Ott 2022 – Mar 2026
University of Milan-Bicocca — *Milan, Italy*

- **Curriculum focus:** Mathematical Analysis, Algebra, Computer Science.
- **Thesis:** *Fondamenti matematici della Crittografia: Dal sistema RSA ai protocolli Quantistici* (Advisor: Prof. Andrea Previtali).
- Analyzed the mathematical foundations of classical cryptography, focusing on the RSA cryptosystem and modular arithmetic.
- Explored quantum cryptography, specifically Quantum Key Distribution (QKD) and the BB84 protocol, as a response to future threats posed by Shor's Algorithm.

Scientific High School Diploma — 2017 – 2022
Istituto Elvetico Salesiani Don Bosco — *Lugano, Switzerland*

ACADEMIC PROJECTS

DiffuBox: Refining 3D Object Detection — 2026
Machine Learning Project

- Implemented a diffusion-based approach for 3D object detection refinement using a domain-agnostic point diffusion model.
- Conditioned the model on LiDAR points to simultaneously refine a bounding box's location, size, and orientation to overcome domain shift.

Bayesian Image Analysis in Fourier Space — 2026
Computational Science Project

- Implemented a Python-based Bayesian framework in Fourier space, leveraging MAP optimization and parallelized MCMC sampling to perform fast, high-accuracy image restoration on microscopy data.

Quantitative Stock Market Analysis — 2025
Python Data Analysis Project

- Developed a quantitative tool in Python to process and analyze historical stock market datasets, leveraging libraries such as NumPy and Pandas.

EXPERIENCE

Private Tutor — 2019 – Present
Self-Employed — *Lugano / Milan*

- Provided tailored private lessons in Mathematics and Physics to middle and high school students.
- Developed customized learning plans to address individual needs, helping students improve their grades and overall subject comprehension.

TECHNICAL SKILLS

- **Programming Languages:** Python (NumPy, Pandas), C/C++, MATLAB, SQL.
- **Software & Tools:** LaTeX, Microsoft Office Suite (Word, Excel, PowerPoint).
- **Theoretical Mathematics:** Mathematical Analysis I & II, Complex Analysis, Linear Algebra, Geometry, Logic.
- **Applied & Computational Math:** Bayesian Inference, Markov Chain Monte Carlo (MCMC), Numerical Optimization, Cryptography.

LANGUAGES

- **Italian:** Native.
- **English:** B2+ Level (IELTS Certificate)
- **German:** B1 Level.